



مدارس
للتكنولوجيا
التطبيقية



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APPLIED TECHNOLOGY
وزارة التربية والتعليم والتعليم الفني



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IT Department
Networks & Cybersecurity



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Python Codes of Automation



Code 1

```
import getpass
import telnetlib

HOST = "192.168.200.254"
user = input("Username: ")
password = getpass.getpass()

tn = telnetlib.Telnet(HOST)

tn.read_until(b"Username: ")
tn.write(user.encode('ascii') + b"\n")
if password:
    tn.read_until(b"Password: ")
    tn.write(password.encode('ascii') + b"\n")

tn.write(b"enable\n")
tn.write(b"we\n")
tn.write(b"conf t\n")
tn.write(b"interface loopback0\n")
tn.write(b"ip address 1.1.1.1 255.255.255.255\n")
tn.write(b"end\n")
tn.write(b"exit\n")

print(tn.read_all().decode('ascii'))
```

Code 2

```
from netmiko import ConnectHandler
# Define device details
R1 = {
    "device_type": "cisco_ios",
    "host": "192.168.200.254",
    "username": "we",
    "password": "we",
    "secret": "we",
}
try:
    net_connect = ConnectHandler(**R1)
    net_connect.enable()
    print(net_connect.send_command("show ip interface
brief"))
    net_connect.disconnect()
except Exception as e:
    print(f"Error: {e}")
```

Code 3

```
import telnetlib
host = "192.168.200.253"
user = "we"
password = "we"
tn = telnetlib.Telnet(host)
tn.read_until(b"Username: ")
tn.write(user.encode('ascii') + b"\n")
if password:
    tn.read_until(b"Password: ")
    tn.write(password.encode('ascii') + b"\n")
tn.write(b"enable\n")
tn.write(b"we\n")
tn.write(b"conf t\n")
tn.write(b"vlan 2\n")
tn.write(b"exit\n")
tn.write(b"vlan 3\n")
tn.write(b"exit\n")
tn.write(b"vlan 4\n")
tn.write(b"exit\n")
tn.write(b"end\n")
tn.write(b"exit\n")
print(tn.read_all().decode('ascii'))
```

Code 4

```
from netmiko import ConnectHandler
# Define device details
devices = [
    {
        "device_type": "cisco_ios",      # IP address of R1
        "host": "192.168.200.254",      # Username for SSH
        "username": "we",               # Password for SSH
        "password": "we",               # Enable secret
        "secret": "we",
    },
    {
        "device_type": "cisco_ios",      # IP address of IOU1
        "host": "192.168.200.253",      # Username for SSH
        "username": "we",
        "password": "we",
        "secret": "we"
    }
]
# Loop through devices and save configurations
for device in devices:
    try:
        # Establish SSH connection
        net_connect = ConnectHandler(**device)
        net_connect.enable()

        # Set terminal length to 0 to disable pagination
        net_connect.send_command("terminal length 0")
        # Get the hostname of the device
        hostname = net_connect.find_prompt().strip("#")
        # Pull running configuration
```

```
config = net_connect.send_command("show
running-config")
# Save configuration to a file named after the
hostname
filename = f"config_{hostname}.txt"
with open(filename, "w") as file:
    file.write(config)
print(f"Configuration saved for {hostname} in
{filename}")
# Disconnect the session
net_connect.disconnect()
except Exception as e:
    print(f"Failed to retrieve configuration from
{device['host']}: {e}")
```

Code 5

```
from netmiko import ConnectHandler
from ipaddress import ip_network
# Define device details
devices = [
```

```

{
    "device_type": "cisco_ios",
    "host": "192.168.200.254",  # R1
    "username": "we",
    "password": "we",
    "secret": "we",  # Enable secret
},
{
    "device_type": "cisco_ios",
    "host": "192.168.200.252",  # R2
    "username": "we",
    "password": "we",
    "secret": "we",
}
]

# OSPF Configuration
ospf_process_id = 1  # OSPF process ID
for device in devices:
    try:
        # Connect to the device
        net_connect = ConnectHandler(**device)
        net_connect.enable()
        # Get the list of interfaces and their subnet masks
        output = net_connect.send_command("show ip interface
brief")
        interfaces = output.splitlines()
        # List to hold OSPF configuration commands
        ospf_commands = [f"router ospf {ospf_process_id}"]
        for line in interfaces[1:]:  # Skip the header line
            if "up" in line and "up" in line.split()[-1]:  #
Check for active interfaces
                parts = line.split()
                ip_address = parts[1]
                if ip_address != "unassigned":

```

```

# Retrieve the subnet mask using "show
run interface <interface>"
    interface_name = parts[0]
    int_output =
net_connect.send_command(f"show run interface
{interface_name}")
    for config_line in
int_output.splitlines():
        if "ip address" in config_line:
            _, ip, subnet_mask =
config_line.split()[-3:]
            # Calculate the network and
wildcard mask
            network =
ip_network(f"{ip}/{subnet_mask}", strict=False)
            ospf_commands.append(
                f"network
{network.network_address} {network.hostmask} area 0"
            )
# Send OSPF configuration commands to the device
net_connect.send_config_set(ospf_commands)
print(f"OSPF configured on {device['host']}:")
for cmd in ospf_commands:
    print(cmd)
# Save configuration
net_connect.save_config()
net_connect.disconnect()
except Exception as e:
    print(f"Failed to configure OSPF on
{device['host']}: {e}")

```


Code 6

```
import getpass
import telnetlib
HOST = "192.168.126.201"
user = input("Username: ")
password = getpass.getpass()
tn = telnetlib.Telnet(HOST)
tn.read_until(b"Username: ")
tn.write(user.encode('ascii') + b"\n")
if password:
    tn.read_until(b"Password: ")
    tn.write(password.encode('ascii') + b"\n")
tn.write(b"enable\n")
tn.write(b"we\n")
tn.write(b"conf t\n")
```

```
for vlan_id in range(2, 12): # This will create VLANs from
2 to 11
    tn.write(f"vlan {vlan_id} \n".encode('ascii')) # Create
VLAN
    tn.write(f"name WE-{vlan_id} \n".encode('ascii')) # Set
VLAN name
tn.write(b"end\n")
tn.write(b"exit\n")
print(tn.read_all().decode('ascii'))
```